

Market Crash Predictability

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Abstract

Unlike Greenwood, Hanson, Shleifer, and Sørensen (2022), who investigate predictable financial crises at the business cycle frequency (one to four years), we examine short-term market crash predictability within a one- to six-month horizon. We find that a combination of widened interest rate differentials and weak stock market performance over the preceding 12 months is associated with a 17.4% probability of a market crash within the next three months. In contrast, during normal periods—when neither interest rate differentials widens nor asset prices experience sharp declines—the probability is approximately 0.1%. Our findings challenge the belief that market crashes are inherently unpredictable. Rather than focusing solely on credit growth or supply dynamics within individual economies as discussed in the literature, we focus on funding costs and the relative attractiveness of investments across markets, exploring how interest rate differentials between local and U.S. rates contribute to cross-country crash prediction.

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1. Introduction

Rather than focusing on business cycle frequency forecasts of financial crises, we prioritize predicting stock market crashes at the monthly level. Additionally, instead of analyzing credit growth or credit supply within individual countries (Borio and Lowe, 2002; Schularick and Taylor, 2012; Krishnamurthy and Muir, 2017; Greenwood, Hanson, Shleifer, and Sørensen, 2022), we assess funding costs in each country and the relative attractiveness of investing in each country’s assets from the cross-country perspective. In international finance, each country’s interest rate differentials can be interpreted as a spread against U.S. interest rates. Consequently, a widened spread could trigger a market crash. Specifically, we investigate how interest rate differentials—which link local and U.S. interest rates—contribute to cross-country crash prediction. This concept aligns with the financial frictions discussed in Hall (2011), where a wedge between borrowing and lending contributes to financial crises, as well as the credit spread theory (Gilchrist and Zakrajšek, 2012; Krishnamurthy and Muir, 2017), which highlights the role of spread changes in shaping economic activity. Our goal is to provide evidence that financial crashes can be effectively predicted using just two country-level variables: changes in interest rate differentials and equity performance.

We first investigate whether stock returns can be predicted using two simple variables: twelve-month changes in interest rate differentials (between the local market and the U.S.) and the twelve-month log change in equity index prices. Our findings indicate that a one-standard-deviation increase in interest rate differential changes is associated with negative returns of -0.8, -2.5, and -4.5 percentage points within one, three, and six months, respectively. Additionally, the R^2 value increases as the horizon extends. Conversely, a one-standard-deviation increase in equity performance does not exhibit significant return predictability.

Next, we examine crash probability. We find that a one-standard-deviation increase in interest rate differential changes corresponds to a 1.7, 3.3, and 3.7 percentage point increase in the probability of a crash occurring within one, three, and six months, respectively. Similarly,

the R^2 value rises with the forecast horizon. Furthermore, a one-standard-deviation increase in equity performance is associated with a 2.8, 3.2, and 2.6 percentage point increase in the probability of a crash occurring within one, three, and six months, respectively.

Using two simple variables—past changes in interest rate differentials and past equity performance—we construct the R-zone. Within this R-Zone, the probability of a market crash over the next three months is approximately 17.4% based on our threshold crash definition, and it rises to 32.0% when using the Greenwood et al. (2022) definition.

Given that interest rate differentials, when measured against the U.S., exhibit a common trend, we examine whether the average R-zone across countries improves our ability to predict market crashes. By constructing a Global R-Zone from both the full sample and an expanding sample, we run panel regressions and our findings reveal that Global R-Zone significantly impact each market’s crash probability.

We proceed to examine the probability of a severe economic contraction and cumulative real economic growth over the next one to six months as a function of past spread changes and equity price fluctuations. Severe economic contraction is defined as an annualized real (log) industrial production growth rate of -2% or lower within specific horizons. Our findings indicate that a combination of widened interest rate differentials and weak stock market performance over the preceding 12 months can predict economic growth. For example, such combination is associated with a 21.5% higher probability of a severe economic contraction within the next three months compared to the median group. Regarding cumulative real economic growth, this same combination corresponds to a 1.7% decline in economic growth over the next three months relative to the median group.

We also evaluate the predictive efficacy and early warning signals of market crashes. On average, we find that a crash can be anticipated approximately seven months in advance. Moreover, adopting looser criteria for widened spreads improves the true positive rate (TPR) and, when a true positive occurs, provides an earlier warning of an impending crisis. However, this comes at the cost of lower specificity (lower TNR) and reduced

precision (lower PPV). Additionally, we examine the economic outcomes associated with false negatives—specifically, R-zone events that did not result in a crash—and find that true positive cases have a greater economic impact than false positives. Extending our analysis to country-level portfolio flows, we observe that widened interest rate differentials predict fund outflows but not weak performance. Meanwhile, true positive cases are associated with negative outflows, aligning with a heightened risk of market crashes.

To gain a clearer understanding of crash predictability, we first decompose changes in interest rate differentials into local interest rate movements and U.S. rate changes. We examine whether each of these components contributes to crash prediction and find that both increased local short-term interest rates and decreased U.S. short-term interest rates elevate the probability of a market crash in each local market.

Additionally, we conduct multiple robustness tests by exploring alternative thresholds for defining crash events. Beyond the standard 4% threshold for each horizon, we also consider 3% and 5% as alternative benchmarks. Furthermore, instead of relying solely on predictors constructed from the past twelve months of data, we extend our analysis to changes over six and eighteen months. Finally, we adopt the crash risk definition used by Greenwood et al. (2022), where a one-month crash event is identified if a market downturn occurs at least once within a given horizon. Given this approach, it is unsurprising that we observe a sharp increase in crash predictability. For instance, a combination of widened interest rate differentials and weak stock market performance over the preceding 12 months corresponds to a 48.3% probability of a market crash within the next six months.

Previous studies have shown that several early warning signals—particularly rapid growth in aggregate credit—can help predict the onset of financial crises. Rather than focusing on annual forecasts, our primary emphasis is on predicting stock market crashes at the monthly level. Furthermore, we examine whether cross-country short-term interest rate changes, alongside the credit growth narrative, could trigger market crashes.

Our research makes several contributions to the existing literature. First, we provide

robust evidence of the interaction between interest rate differentials and equity performance, utilizing a simple yet transparent methodology. The R-Zone effect on potential crashes is substantial, with the Global R-Zone accounting for a significant portion of crash probability. This underscores the importance of relative funding costs and asset attractiveness across countries. Our findings offer fresh insights that contribute to the global financial cycle literature (Rey (2015), Miranda-Agrippino and Rey (2020), etc.). Additionally, our analysis of crash probability enriches the global disaster measures literature (Manela and Moreira, 2017; Huang and Kilic, 2019; Chen, Yao, Zhang, and Zhu, 2023; Goetzmann, Kim, and Shiller, 2022). Second, we identify a higher degree of market crash predictability under both threshold definitions and the Greenwood et al. (2022) crash framework, with effect sizes exceeding those in prior studies. This suggests that short-term market crashes are, to some extent, predictable. Moreover, such predictability is closely linked to deteriorating macroeconomic outcomes (Baron and Xiong, 2017; López-Salido, Stein, and Zakrajšek, 2017; Mian, Sufi, and Verner, 2017; Krishnamurthy and Muir, 2017; Richter, Schularick, and Wachtel, 2021; Greenwood et al., 2022; Kirti, 2025). Finally, we demonstrate that both rising local interest rates and declining U.S. interest rates significantly increase the probability of a market crash. These findings provide new empirical evidence on how changes in interest rates in the U.S. and non-U.S. economies contribute to crash risk (Rey (2015), Miranda-Agrippino and Rey (2020), etc.).

Our findings are consistent with the Minsky (1977, 1986) and Kindleberger (1970) framework on credit cycles and financial crises. Recent theoretical models emphasize the role of expectations in shaping credit supply and borrowing dynamics. Excessive credit expansion, often fueled by overly optimistic beliefs, lays the groundwork for future financial crises (Bordalo, Gennaioli, and Shleifer, 2018; Gennaioli and Shleifer, 2018; Greenwood, Hanson, and Jin, 2019; Maxted, 2024; Krishnamurthy and Li, 2025). In particular, they highlight how shifts in credit supply can amplify economic downturns and contribute to financial instability (Hall, 2011; Eggertsson and Krugman, 2012; Gilchrist and Zakrajšek, 2012; He and

Krishnamurthy, 2013; Korinek and Simsek, 2016; Guerrieri and Lorenzoni, 2017; Krishnamurthy and Muir, 2017; Bordalo, Gennaioli, Shleifer, and Terry, 2021; Greenwood et al., 2022). Rather than concentrating solely on credit growth or credit supply dynamics within individual economies as discussed in the literature, we shift the focus to funding costs and the relative attractiveness of investments across different markets. Specifically, we investigate how interest rate differentials—linking local and U.S. interest rates—contribute to cross-country crash prediction, demonstrating how monetary policy coordination can elevate the risk of stock market crashes.

Our paper contributes to the discussion on global financial cycles and U.S. monetary policy, particularly in their role in shaping global financial markets. Previous studies, such as (Rey, 2015; Miranda-Agrippino and Rey, 2020, 2022; Brusa, Savor, and Wilson, 2020; Cieslak, Morse, and Vissing-Jorgensen, 2019), have examined how the Federal Reserve, as the global central bank, and U.S. monetary policy influence global stock returns. Our research extends this literature by focusing on crash risk. We document that tightening local credit conditions or raising short-term interest rates can increase crash risk. However, we also find that a decrease in U.S. short-term interest rates similarly contributes to a higher crash probability. Thus, we provide complementary evidence on how an increasing spread between local and U.S. interest rates affects financial stability—whether due to higher local rates or an intentional decrease in U.S. interest rates. Complementing Miranda-Agrippino and Rey (2020)’s finding, we highlight that a decrease in U.S. interest rates can lead to a market crash, suggesting a different stage in U.S. monetary policy decision-making.

The rest of the paper is organized as follows. In the next section, we introduce the data and standard framework for documenting crash predictability and present the R-Zone’s role in predicting market crashes. Section 3 explores the distinction between Local R-Zone and Global R-Zone. Section 4 investigates whether movements in interest rate differentials are associated with economic growth. Section 5 examines early warning signals and fund flows related to market crashes. Section 6 extends various robustness tests, and Section 7

concludes.

2. Predicting Market Crashes

2.1. Data

Our monthly data consist of indicator variables for financial crashes, and interest rate differentials, and equity prices, which we collect for 35 countries from 1995 through 2019.

The key dependent variables in most of our analysis are binary indicators for the onset of a financial crash. Following previous literature, we assume that crises account for approximately 3% - 4% of the sample and define a market crash as the bottom 4% of stock returns.¹ The main difference is that financial crises are typically measured at an annual frequency, whereas here, we define market crashes at the monthly, quarterly, and half-year levels.

In robustness tests, we also: (1) set 3% or 5% as the threshold for defining a crash, (2) apply expanding window estimation to dynamically redefine the 4% crash threshold at each step, and (3) define the crash exactly as in Greenwood et al. (2022). The observed patterns remain robust across different crash specifications.

As shown in Table 1, we report the threshold (TH) corresponding to the bottom 4% of stock returns for each horizon. Additionally, we provide the mean and standard deviation of returns below the threshold. Specifically, the threshold for entering a crash region occurs when stock prices drop by 11.57%, 20.49%, or 29.55% over one-, three-, and six-month horizons.

[Insert Table 1 near here]

Our sample is an unbalanced panel from 35 countries (markets) over the period 1995 to 2019. It includes 23 advanced markets and 12 emerging markets. The advanced markets are

¹According to Greenwood et al. (2022)'s summary, the unconditional probability of a crisis onset in any given country-year is 4.0%, based on Baron, Verner, and Xiong (2021). The Jordà, Schularick, and Taylor (2017) indicator estimates an unconditional probability of 2.6%, while Reinhart and Rogoff (2011) indicator shows a probability of 3.6%.

Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Hong Kong, Ireland, Italy, Japan, the Netherlands, New Zealand, Norway, Portugal, Singapore, Spain, Sweden, Switzerland, the UK, and the US. The emerging markets are Brazil, Chile, China, the Czech Republic, India, Malaysia, Mexico, the Philippines, Poland, South Africa, South Korea, and Thailand. Our framework allows for the incorporation of more countries, provided there is reliable data on short-term interest rates.

The major data explored in this paper are interest rate differential changes, which rely on the short-term interest rates of each country and the stock prices of each market. Equity price indices (P) are retrieved from MSCI database. Short-term interest rates are retrieved primarily from OECD statistics, supplemented with deposit rate data from IMF’s International Financial Statistics (IFS) database.² As shown in Table 1, *Spread* refers to the interest rate differentials against the U.S. Δ_{12} denotes changes over twelve months. Besides that, industrial production growth are retrieved primarily from OECD statistics, supplemented with GDP growth data from World Bank’s Global Economic Monitor (GEM) database.

Our data on interest rate differentials and stock prices are summarized in the bottom panel of Table 1. Our baseline sample includes every country-month observation beginning in 1995 and ending in 2019 for which we have data on past twelve-month interest rate differentials changes and equity price changes, as well as the crash indicator in the following six months. The result is an unbalanced panel data set that covers 35 countries.

2.2. *Predicting Market Returns with Interest Rate Differentials*

Before analyzing crash predictability, we first examine whether stock returns can be predicted. Specifically, we assess whether interest rate differentials or equity prices contribute to return forecasting.

²The exception is Brazil, where we use the short-term interest rate from the Brazilian central bank.

Table 2 presents linear forecasting regressions of the form

$$Ret_{i, t+1 \text{ to } t+h} = \alpha_i^h + \beta^h \cdot \Delta_{12}X_{i,t} + \epsilon_{i, t+1 \text{ to } t+h} \quad (1)$$

$h = 1, 3$, and 6 , where α_i^h is a country fixed effect, Δ_{12} is the change in predictor $X_{i,t}$ over the twelve months ending in t , and $Ret_{i, t+1 \text{ to } t+h}$ is stock returns in country i , spanning from month $t+1$ to month $t+h$. To draw appropriate statistical inferences, we follow Greenwood et al. (2022) and compute t-statistics (shown in square brackets) using Driscoll and Kraay (1998) standard errors.

As predictors, we examine twelve-month changes in interest rate differentials (between the local and the U.S.), and the twelve-month log change in equity index price.³ Each of these variables is normalized by its sample standard deviation, so the coefficient shows corresponding stock returns within h months if past twelve-month predictor X rises by one standard deviation.

[Insert Table 2 near here]

Table 2 shows that interest rate differential changes forecasts stock returns in each country. As shown in columns 1-3, a one-standard-deviation increase in $\Delta_{12}Spread$ is associated with a -0.8, -2.5, and -4.5 percentage point negative returns within one, three, and six months, respectively. The R^2 value also increases with the horizon. $\Delta_{12}log(P)$ does not exhibit significant predictability according to columns 4-6.

³We also examine the six-month and eighteen-month changes for each predictor and report the results in the appendix.

2.3. Predicting Market Crashes with Interest Rate Differentials and Asset Price Growth

We proceed to examine the crash probability and Table 3 presents linear forecasting regressions of the form

$$Crash_{i, t+1 \text{ to } t+h} = \alpha_i^h + \beta^h \cdot \Delta_{12}X_{i,t} + \epsilon_{i, t+1 \text{ to } t+h} \quad (2)$$

$h = 1, 3$, and 6 , where α_i^h is a country fixed effect, Δ_{12} is the change in predictor $X_{i,t}$ over the twelve months ending in t , and $Crash_{i, t+1 \text{ to } t+h}$ is an indicator variable equal to one if a crash begins in country i in month between $t + 1$ and month $t + h$ — i.e., defining $Crash-Start_{i, t}$ as an indicator that switches on if a crash begins in country i in month t , we define a market crash at horizons of one, three, and six months, respectively. In Section 5.3, we also report the results where defining crisis following Greenwood, Hanson, Shleifer, and Sørensen (2022) that $Crash_{i, t+1 \text{ to } t+h} = \max\{Crash-Start_{i, t+1}, \dots, Crash-Start_{i, t+h}\}$. To draw appropriate statistical inferences, we follow Greenwood et al. (2022) and compute t-statistics (shown in square brackets) using Driscoll-Kraay (1998) standard errors.

As predictors, we also examine twelve-month changes in interest rate differentials (between the local and the U.S.), and the twelve-month log change in equity index price. Each of these variables is normalized by its sample standard deviation, so the coefficient gives the change in the probability of a crash beginning within h months if past twelve-month predictor X rises by one standard deviation.

[Insert Table 3 near here]

Table 3 shows that interest rate differential changes forecasts the probability of a crash in each country. As shown in columns 1-3, a one-standard-deviation increase in $\Delta_{12}Spread$ is associated with a 1.7, 3.3, and 3.7 percentage point increase in the probability of a crash beginning within one, three, and six months, respectively. Both the coefficients and R^2

value increases with the horizon. $\Delta_{12}\log(P)$ also exhibit significant predictability according to columns 4-6. A one-standard-deviation decrease in $\Delta_{12}\log(P)$ is associated with a 2.8, 3.2, and 2.6 percentage point increase in the probability of a crash beginning within one, three, and six months, respectively.

2.4. *Market Crashes Probability Distribution and R-Zone Construction*

In this section, motivated by prior work suggesting that augmenting asset price changes could improve crisis prediction (Borio and Lowe (2002), Borio and Drehmann (2009), and Greenwood et al. (2022)), we explore whether market crash can be better foretasted by jointly examining the interest rate differentials and the price changes.

We follow Greenwood et al. (2022) by dividing all country-months in our sample into 15 bins based on past price change tercile and past interest rate differentials change quintile. The assignment thresholds are based on the distribution of two predictors in our full panel data set and thus are the same for all 35 markets in the sample. We then compute the probability of a crash beginning within the next h months conditional on being in each bin. In Table 4, we show a nonparametric way of understanding the multivariate nonlinear relationship between past interest rate differentials and asset price change and the probability of a future crash at various horizons h .

[Insert Table 4 near here]

In Panel B, Table 4 the first matrix on the left reports the probability of a crisis arriving within one month based on past interest rate differentials and equity prices. The Spread Quintile (1 to 5) reflects the widening of interest rate differentials, while the Price Tercile (1 to 3) represents the equity performance from strong to weak. For example, when the spread is narrow and stock performance is strong (Spread Quintile = 1, Price Tercile = 1), the probability of a crash occurring within one month is 3.1%. Conversely, when the spread is widened and stock performance is weak (Spread Quintile = 5, Price Tercile = 3), the crash

probability increases to 14.1%. The matrix on the right reports the difference between the conditional probability for each bin and the probability for the “median” bin where price growth is in the middle tercile and interest rate differentials change is the third quintile. We find the previous crash probability is 13.4% and this difference in probabilities is statistically distinguishable from zero.

Conditional on a widened spread and weak stock performance, the cumulative probability of a crash increases sharply as the forecast horizon extends. A combination of widened interest rate differentials and weak stock market performance over the preceding 12 months is associated with a 17.4% probability of a market crash within the next three months. In contrast, during normal periods—when neither interest rate differentials widen nor asset prices experience sharp declines—the probability is approximately 0.1%.

To explore crash prediction in detail, we define the three indicator variables

$$Widened-Spread_{i,t} = \mathbf{1}\{\Delta_{12}Spread_{i,t} > 80^{th}percentile\} \quad (3)$$

$$Weak-Performance_{i,t} = \mathbf{1}\{\Delta_{12}\log(P)_{i,t} < 33.3^{th}percentile\} \quad (4)$$

$$R-Zone_{i,t} = Widened-Spread_{i,t} \times Weak-Performance_{i,t} \quad (5)$$

where the cutoffs are based on the distribution of spread changes and price growth in our full country-month panel. The Red zone (R-zone) is the interaction between these two indicators, so it only switches on when widened spread and weak performance are jointly elevated.

To assess how widened spread and weak performance jointly affect the probability of a future crash, in Table 5, we run the following regression

$$Crash_{i, t+1 \text{ to } t+h} = \alpha_i^h + \beta^h \cdot Widened-Spread_{i,t} + \delta^h \cdot Weak-Performance_{i,t} + \gamma^h \cdot R-Zone_{i,t} + \epsilon_{i, t+1 \text{ to } t+h} \quad (6)$$

h = 1, 3, and 6, where $Crash_{i, t+1 \text{ to } t+h}$ is defined as above. The sum of the coefficients

$\beta^h + \delta^h + \gamma^h$ gives the increase in the probability that a crisis begins within h months when widened spread and weak stock performance are jointly elevated.

Compared to the findings reported in Table 3, these predictive regressions allow us to separately estimate the direct relationship between widened spread and weak performance and the future probability of a crash, as well as their interaction, R-zone. We include a full set of country fixed effects α^h to focus on within country time-series variation, however, but we obtain very similar results in Table 5 if we omit the country fixed effects.

[Insert Table 5 near here]

Table 5 presents the results. Conditional on entering the R-zone, the cumulative probability that a market crash arrives increases for the first six months. The coefficient on the R-zone interaction term is economically large and statistically significant at all prediction horizons. We also examine whether using only the R-zone indicator is sufficient. Comparing the full specifications, we find that in the model including only the R-zone interaction effect at each horizon, there is little loss in forecasting ability in terms of the coefficients and the R^2 .

[Insert Table 6 near here]

In Table 6, we further provide robust evidence by examining the predictability of advanced financial markets and emerging markets. We find consistent evidence that R-Zones are significant across all horizons, with forecasting power being stronger for advanced financial markets in terms of the coefficients and R^2 , especially as the horizon expands. To ensure no look-ahead bias, we also investigate the predictability of the R-Zone by reestimating it using an expanding sample each month. Using the expanding R-Zone, we document consistent evidence of elevated crash risk.

3. Local versus Global R-Zones

Given that all interest rate differentials are measured against the U.S., there could be a common trend driving changes over time. Therefore, in Table 7 we examine whether the average R-zone across countries improve our ability to predict crashes. Using the country-month panel, we estimate regressions of the form

$$Crash_{i, t+1 \text{ to } t+h} = \alpha_i^h + \gamma^h \cdot Local\ R\text{-}Zone_{i,t} + \xi^h \cdot Global\ R\text{-}Zone_t + \epsilon_{i, t+1 \text{ to } t+h} \quad (7)$$

$h = 1, 3$, and 6 , where $Crash_{i, t+1 \text{ to } t+h}$ is defined as above. We report results using the R-Zone using the full sample and from the expanding method. The two global R-Zones are shown in Figure 1.

[Insert Figure 1 near here]

In Table 7, we find that both the Local R-Zone and Global R-Zone predict higher crash risk in Panel A, while only the Global R-Zone remains significant when using the expanding measures, according to Panel B. Results suggest that changes in interest rate differentials share a global component and have a significant effect on each market’s crash probability.

[Insert Table 7 near here]

4. Interest Rate Differentials Movements and Future Economic Growth

Economists have long recognized that the aftermath of a financial crisis is usually marked by a significant decline in real economic activity. (Kaminsky and Reinhart (1999), Reinhart and Rogoff (2009), and Cecchetti, Kohler, and Upper (2009)). Greenwood et al. (2022) also document such pattern follows the high debt. Given our focus on short-term crash

probability, our evidence on crash-related economic impact could offer novel insights. To measure this impact, we select industrial production growth as our key indicator.

[Insert Table 8 near here]

In Table 8, we examine the probability of a severe economic contraction⁴ and cumulative real economic growth within the next one to six months as a function of past spread changes and past equity price changes. A combination of widened interest rate differentials and weak stock market performance over the preceding 12 months is associated with a 21.5% higher probability of a severe economic contraction within the next three months compared to the median group. Regarding cumulative real economic growth, a combination of widened interest rate differentials and weak stock market performance over the preceding 12 months is associated with a 1.7% decline in economic growth within the next three months compared to the median group.

5. Crash Prediction, Early Warning, and Fund Flows

We follow Greenwood et al. (2022) to assess the predictive efficacy and early warning signals. As shown in Table 9, we examine the “precision” or positive predictive value (PPV), the “sensitivity” or true positive rate (TPR), and the “specificity” or true negative rate (TNR) of R-zone. Besides the R-Zone, we further examine the Y-Zone by relaxing the conditions.

[Insert Table 9 near here]

On average, we find that a market crash can be anticipated approximately seven months in advance. Furthermore, adopting looser criteria for widened spreads can increase the true positive rate (TPR) and, conditional on a true positive, provide earlier warning of an

⁴We define severe economic contraction as experiencing annualized real (log) industrial production growth of -2% or less within specific horizons.

impending crisis. However, this comes at the cost of lower specificity (lower TNR) and, consequently, reduced precision (lower PPV).

In Table 10, we further examine the economic outcomes associated with false negatives, specifically R-zone events that did not lead to a crash. We estimate regressions of the following form

$$\Delta \log(IP)_{i, t+1 \text{ to } t+h} = \alpha_i^h + \gamma^{h, tp} \cdot R\text{-}Zone_{i,t} \cdot Crash_{i, t+1 \text{ to } t+6} + \gamma^{h, fp} \cdot R\text{-}Zone_{i,t} \cdot (1 - Crash_{i, t+1 \text{ to } t+6}) + \epsilon_{i, t+1 \text{ to } t+h} \quad (8)$$

$h = 1, 3$, and 6 , where $\Delta \log(IP)_{i, t+1 \text{ to } t+h}$ is the log IP growth in country i from month $t + 1$ to month $t + h$. $Crash_{i, t+1 \text{ to } t+h}$ is defined as above.

We find that both $\gamma^{h, tp}$ and $\gamma^{h, fp}$ are negative; however, the magnitudes of $\gamma^{h, tp}$ are larger than those of $\gamma^{h, fp}$, and the difference increases with the forecast horizon. This suggests that true positive cases have a greater economic impact than false positive cases.

[Insert Table 10 near here]

To further investigate the relationship between changes in interest rate differentials and fund flows in each market related to index crash risk, we collect investment portfolio data for analysis. We collect data from IMF - Coordinated Portfolio Investment Survey and investigate the position changes. We estimate regressions of the following form

$$Fund\text{-}Flow_{i, t+1 \text{ to } t+h} = \alpha_i^h + \gamma^{h, tp} \cdot R\text{-}Zone_{i,t} \cdot Crash_{i, t+1 \text{ to } t+6} + \gamma^{h, fp} \cdot R\text{-}Zone_{i,t} \cdot (1 - Crash_{i, t+1 \text{ to } t+6}) + \epsilon_{i, t+1 \text{ to } t+h} \quad (9)$$

$h = 1, 3$, and 6 , where $Fund\text{-}Flow_{i, t+1 \text{ to } t+h}$ is the change of reported portfolio investment assets in country i from month $t + 1$ to month $t + h$. $Crash_{i, t+1 \text{ to } t+h}$ is defined as above.

[Insert Table 11 near here]

In Table 11, we find that widened interest rate differentials predict fund outflows but

not weak performance. Meanwhile, the true positive case ($\gamma^{h,tp}$) predicts negative outflows, whereas the false positive case ($\gamma^{h,fp}$) leads to fund inflows on a smaller scale. The signs of coefficients β and $\gamma^{h,tp}$ indicate fund flows that align with high crash risk.

6. Understanding Crash Predictability - More Results and Robustness

In this section, we first examine whether the effects of interest rate differential changes are driven by local interest rate movements or U.S. rate changes (Interest Rate Differentials Decomposition). For robustness checks, we investigate alternative thresholds for defining crash events and different horizons for constructing predictors (Robust: Alternative Crash Threshold and Horizons). Lastly, we follow Greenwood et al. (2022) in defining crash risk, where a one-month crash event occurs at least once within a certain horizon (Crisis Definition Following Greenwood, Hanson, Shleifer, and Sørensen (2022)). All related results are reported in the appendix.

6.1. *Interest Rate Differentials Decompositions: Local vs US Short-term Interest rate changes*

If we decompose the interest rate differential changes, we will have the local interest rate movements and U.S. rate changes ($\Delta_{12}Spread = \Delta_{12}st_{Local} - \Delta_{12}st_{US}$). Tables A1-A4 show both local interest rate movements or U.S. rate changes contribute to the high market crash risk. Results suggest that both increased local short-term interest rates and decreased U.S. short-term interest rates can contribute to a higher market crash in each local market.

6.2. *Robust: Alternative Crash Threshold and Horizons*

We investigate alternative thresholds for defining crash events. In the main analysis, we define a crash event as returns dropping below 4% for each horizon. Here, we further examine alternative thresholds by setting them at 3% or 5%. Crash predictability remains robust across different crash conditions. Relaxing the crash conditions increase the crash predictability based on the increased R^2 . Additionally, we explore different horizons for constructing predictors by analyzing changes over six or eighteen months. Predictors that incorporate changes over different horizons exhibit consistent predictive power. Tables A5-A7 show robust patterns.

6.3. *Crisis Definition Following Greenwood, Hanson, Shleifer, and Sørensen (2022)*

Lastly, we follow Greenwood et al. (2022) in defining crash risk, where a one-month crash event occurs at least once within a given horizon. We reproduce the results and report them in Tables A8–A11. Since Greenwood et al. (2022) measure the probability of a crash occurring at least once within a certain horizon, it is unsurprising that we document a sharp increase in crash predictability. For example, a combination of widened interest rate differentials and weak stock market performance over the preceding 12 months is associated with a 48.3% probability of a market crash within the next six months. In contrast, during normal periods—when neither interest rate differentials widen nor asset prices experience sharp declines—the probability is approximately 3.0%.

7. Conclusion

Using two simple variables—past changes in interest rate differentials and past equity performance—we construct the R-zone. Within this zone, the probability of a market crash over the next three months is approximately 17.4% based on our threshold crash definition,

and it rises to 32.0% when using the Greenwood et al. (2022) definition. Further evidence, by decomposing changes in interest rate differentials into changes in local market short-term interest rates and U.S. short-term interest rates, reveals that both an increase in local interest rates and a decrease in U.S. interest rates contribute to a higher probability of a market crash.

Our conclusion is that financial crashes can be sufficiently predicted using just two country-level variables—changes in interest rate differentials and equity performance. Rather than focusing on annual forecasts of financial crises, our primary emphasis is on predicting stock market crashes at the monthly level. Furthermore, instead of examining credit growth within individual countries, we explore how interest rate differentials—which link local and U.S. interest rates—contribute to cross-country crash prediction, highlighting how monetary policy coordination may elevate the risk of a stock market crash.

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Table 1: Summary Statistics

This table presents summary statistics for our main variables in percent. Our sample is an unbalanced panel from 35 countries over the period 1995 to 2019. Equity price indices (P) are retrieved from MSCI. Short-term interest rates are retrieved primarily from OECD statistics, supplemented with data from IMF-IFS. *Spread* refers to the interest rate differentials against the U.S. Δ_{12} denotes changes over twelve months. A Market Crash is defined as the bottom 4% of the return sample, with the threshold (TH) reported for each horizon.

	N	Mean	Std	TH	Quantiles			
Crashes (Ret<TH)								
Ret-1M(%)	417	-16.78	6.64	-11.57				
Ret-3M(%)	418	-28.92	12.14	-20.49				
Ret-6M(%)	418	-40.08	17.22	-29.55				
$12 * \Delta_{12} Spread(\%)$					Q20	Q40	Q60	Q80
	10,009	-0.22	1.95		-1.09	-0.47	0.00	0.73
$\Delta_{12} log(P)(\%)$					Q33.3	Q66.7		
	10,009	0.20	26.66		-6.06	11.87		

Table 2: Linear Regression - Returns

This table presents results of the regression model:

$$Ret_{i, t+1 \text{ to } t+h} = \alpha_i^h + \beta^h \cdot \Delta_{12}X_{i,t} + \epsilon_{i, t+1 \text{ to } t+h}$$

$h = 1, 3$, and 6 , where α_i^h is a country fixed effect, Δ_{12} is the change in predictor $X_{i,t}$ over the twelve months ending in t , and $Ret_{i, t+1 \text{ to } t+h}$ is stock returns in country i , spanning from month $t + 1$ to month $t + h$. t-statistics are reported in brackets and based on Driscoll and Kraay (1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively, using Kiefer and Vogelsang(2005) corrected p-values. Reported coefficients and R^2 s are in percent.

	Ret_{t+1}	Ret_{t+3}	Ret_{t+6}	Ret_{t+1}	Ret_{t+3}	Ret_{t+6}
$\Delta_{12}Spread$	-0.8*** [-3.64]	-2.5*** [-2.83]	-4.5*** [-2.66]			
$\Delta_{12}log(P)$				0.4 [1.31]	0.8 [0.79]	0.8 [0.50]
$R^2(within)$	1.5	4.2	6.2	0.4	0.4	0.2
N	10,009	10,009	10,009	10,009	10,009	10,009

Table 3: Linear Regression - Crashes

This table presents results of the regression model:

$$Crash_{i, t+1 \text{ to } t+h} = \alpha_i^h + \beta^h \cdot \Delta_{12}X_{i,t} + \epsilon_{i, t+1 \text{ to } t+h}$$

$h = 1, 3$, and 6 , where α_i^h is a country fixed effect, Δ_{12} is the change in predictor $X_{i,t}$ over the twelve months ending in t , and $Crash_{i, t+1 \text{ to } t+h}$ is an indicator variable equal to one if a crash begins in country i in month between $t + 1$ and month $t + h$. t-statistics are reported in brackets and based on Driscoll and Kraay (1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively, using Kiefer and Vogelsang(2005) corrected p-values. Reported coefficients and R^2 s are in percent.

	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}
$\Delta_{12}Spread$	1.7*** [2.59]	3.3** [2.23]	3.7 [1.62]			
$\Delta_{12}log(P)$				-2.8*** [-3.72]	-3.2*** [-2.82]	-2.6** [-2.22]
$R^2(within)$	0.8	2.7	3.4	2.0	2.7	1.7
N	10,009	10,009	10,009	10,009	10,009	10,009

Table 4: Crashes Probabilities by Price and Spread Change Quantiles

Panel A presents the distribution of country-months across price growth terciles and spread change quintiles. Panel B presents the probability of a crash within one to six months for the intersections of the price terciles and spread change quintiles. It also presents the difference in future crash probability between each group and the median group, which is defined as the intersection of the second price tercile and the third spread change quintile. Both price growth and spread change are measured over twelve months. p-values are based on Driscoll and Kraay(1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively, and corrected according to Kiefer and Vogelsang (2005). *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Panel A: Distribution of Observations(%)											
		Spread Quintile									
Price Tercile		1	2	3	4	5					
1		9.3	6.6	6.7	7.1	3.6					
2		5.5	8.1	7.6	7.5	4.7					
3		5.2	5.3	5.7	5.4	11.7					
Panel B: Crash Probabilities(%) by Spread and Equity Returns											
1-month horizon											
		Crash Frequency					Diff. from Median				
		Spread Quintile					Spread Quintile				
Price Tercile		1	2	3	4	5	1	2	3	4	5
	1	3.1	1.2	1.8	2.1	4.2	2.5**	0.5	1.1*	1.4	3.5**
	2	1.8	0.7	0.7	1.1	5.1	1.1*	0.1	0.0	0.4	4.5**
	3	7.8	3.2	3.7	4.2	14.1	7.2***	2.5*	3.0***	3.6***	13.4***
3-month horizon											
		Crash Frequency					Diff. from Median				
		Spread Quintile					Spread Quintile				
Price Tercile		1	2	3	4	5	1	2	3	4	5
	1	1.1	0.6	1.6	2.0	6.9	0.9**	0.5	1.5***	1.8	6.8***
	2	1.8	0.2	0.1	0.5	6.6	1.7**	0.1	0.0	0.4	6.5***
	3	5.9	2.6	4.0	5.0	17.4	5.8**	2.5***	3.9***	4.8***	17.3***
6-month horizon											
		Crash Frequency					Diff. from Median				
		Spread Quintile					Spread Quintile				
Price Tercile		1	2	3	4	5	1	2	3	4	5
	1	2.1	0.3	1.8	1.1	6.6	2.0*	0.2	1.6**	1.0**	6.5**
	2	2.0	0.4	0.1	1.5	7.7	1.8**	0.2	0.0	1.3**	7.6*
	3	6.7	0.9	3.0	3.9	17.9	6.6**	0.8	2.9**	3.7**	17.8***

Table 5: Crashes Prediction with Spread Change and Stock Performance
This table presents results of the regression model

$$Crash_{i, t+1 \text{ to } t+h} = \alpha_i^h + \beta^h \cdot Widened-Spread_{i,t} + \delta^h \cdot Weak-Performance_{i,t} + \gamma^h \cdot R-Zone_{i,t} + \epsilon_i^h$$

$h = 1, 3, \text{ and } 6$. $Widened-Spread_{i,t} \equiv \mathbf{1}\{\Delta_{12}Spread_{i,t} > 80^{th}percentile\}$ is an indicator variable that takes the value of one if twelve-month spread change is in the highest quintile, and $Weak-Performance_{i,t} \equiv \mathbf{1}\{\Delta_{12}log(P)_{i,t} < 33.3^{th}percentile\}$ is an indicator variable that takes the value of one if equity performance is in its lowest tercile, an R-zone is the intersection of high spread change and low stock performance $R-Zone_{i,t} \equiv Widened-Spread_{i,t} \times Weak-Performance_{i,t}$. The sum of the coefficients $\beta^h + \delta^h + \gamma^h$ gives the increase in the probability that a crisis begins within h months when widened spread and weak stock performance are jointly elevated. t-statistics are reported in brackets and based on Driscoll and Kraay (1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively, using Kiefer and Vogelsang(2005) corrected p-values. Reported coefficients and R^2 s are in percent.

	Crash _{t+1}				Crash _{t+3}				Crash _{t+6}			
<i>Widened-Spread</i> (β)	7.8*** [3.72]		3.1* [1.92]		11.2*** [2.87]		5.8*** [3.26]		11.8** [2.07]		6.1 [1.61]	
<i>Weak-Performance</i> (δ)		5.6*** [4.12]	2.6*** [3.92]			6.8*** [2.69]	2.8*** [3.33]			6.3** [2.17]	1.8* [1.92]	
<i>R-Zone</i> (γ)			6.7** [2.21]	11.3*** [3.73]			7.7* [1.64]	15.0*** [2.77]			8.7** [2.32]	15.6** [2.38]
Sum of coefficients($\beta + \delta + \gamma$)			12.3***				16.3***				16.6**	
t-statistic($\beta + \delta + \gamma$)			[3.94]				[2.92]				[2.39]	
R^2 (within)	2.5	1.8	3.9	3.5	5.1	2.6	6.8	6.0	5.6	2.2	7.1	6.4

Table 6: Crashes Prediction with R-zones: Robust

This table presents results of the regression model

$$Crash_{i, t+1 \text{ to } t+h} = \alpha_i^h + \gamma^h \cdot R\text{-Zone}_{i,t} + \epsilon_{i, t+1 \text{ to } t+h}$$

$h = 1, 3$, and 6 . An R-zone is the intersection of high spread change and low stock performance $R\text{-Zone}_{i,t} \equiv \mathbf{1}\{\Delta_{12}Spread_{i,t} > 80^{th}percentile\} \times \mathbf{1}\{\Delta_{12}log(P)_{i,t} < 33.3^{th}percentile\}$. Besides examining advanced market and emerging market, we also investigate the expanding sample evidence. In expanding sample specification, R-zone indicators in each month t are based on a expanding sample using past data, that is, the R-zone indicator in 2006m12 is based on data from 1995m1 to 2006m12, and we require at least 60 months of data. t-statistics are reported in brackets and based on Driscoll and Kraay (1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively, using Kiefer and Vogelsang(2005) corrected p-values. Reported coefficients and R^2 s are in percent.

	Crash _{t+1}			Crash _{t+3}			Crash _{t+6}		
	ADV	EM	Expanding	ADV	EM	Expanding	ADV	EM	Expanding
<i>R-Zone</i>	12.0*** [3.44]	10.3*** [3.44]	10.2** [2.43]	16.1*** [2.64]	13.2*** [2.76]	16.0** [1.96]	18.8** [2.26]	10.5** [2.48]	19.2* [1.82]
$R^2(\text{within})$	4.6	2.3	2.5	7.9	3.8	6.1	10.5	2.4	8.3
N	6,565	3,444	8,365	6,565	3,444	8,365	6,565	3,444	8,365

Table 7: Crashes Prediction with Global R-zones

This table presents results of the regression model

$$Crash_{i, t+1 \text{ to } t+h} = \alpha_i^h + \gamma^h \cdot Local\ R\text{-}Zone_{i,t} + \xi^h \cdot Global\ R\text{-}Zone_t + \epsilon_{i, t+1 \text{ to } t+h}$$

$h = 1, 3, \text{ and } 6$. *Global R-Zone_t* measures the fraction of countries in the red zone at a given month. Panel A presents the standard results, while Panel B reports the results using the R-zones from the expanding sample. t-statistics are reported in brackets and based on Driscoll and Kraay (1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively, using Kiefer and Vogelsang(2005) corrected p-values. Reported coefficients and R^2 s are in percent.

Panel A: R-Zone						
	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}
Local <i>R-Zone</i>	4.3*** [2.80]	5.4** [2.17]	4.4* [1.78]			
Global <i>R-Zone</i>	15.3** [2.55]	20.9* [1.95]	24.4 [1.60]	19.7*** [3.28]	26.3** [2.51]	28.8** [2.04]
$R^2(\text{within})$	5.1	8.8	10.3	4.8	8.4	10.0
N	10,009	10,009	10,009	10,009	10,009	10,009
Panel B: R-Zone (Expanding)						
	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}
Local <i>R-Zone</i>	0.9 [0.52]	1.8 [1.53]	0.5 [0.38]			
Global <i>R-Zone</i>	17.5** [2.32]	26.7* [1.93]	35.1* [1.77]	18.4** [2.47]	28.6** [2.08]	35.7* [1.90]
$R^2(\text{within})$	4.4	10.3	15.2	4.4	10.3	15.2
N	8,365	8,365	8,365	8,365	8,365	8,365

Table 8: Economic Decline Probability and Cumulative IP Growth

Panel A presents the difference in the probability of experiencing annualized real (log) industrial production growth of -2% or less within specific horizons between each group and the median group, which is defined as the intersection of the second price tercile and the third spread change quintile. Panel B presents the difference in future cumulative real (log) industrial production growth between each group and the median group, which is defined as the intersection of the second price tercile and the third spread change quintile. Both price growth and spread change are measured over twelve months. p-values are based on Driscoll and Kraay(1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively, and corrected according to Kiefer and Vogelsang (2005). *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

1-month horizon (Diff. from Median)										
Price Tercile	Econ Decline Prob. Spread Quintile					Cumulative Econ growth Spread Quintile				
	1	2	3	4	5	1	2	3	4	5
1	-0.7	0.0	-2.3	-1.1	-0.4	0.3*	0.1	0.1	0.1	0.3*
2	0.4	0.0	0.0	0.5	2.3	-0.1	0.0	0.0	-0.1	0.0
3	2.1	8.8***	5.9**	7.1**	12.8***	0.0	-0.3**	-0.1	-0.1	-0.7***
3-month horizon (Diff. from Median)										
Price Tercile	Econ Decline Prob. Spread Quintile					Cumulative Econ growth Spread Quintile				
	1	2	3	4	5	1	2	3	4	5
1	-6.8**	-4.7	-5.8**	-1.8	-0.5	0.7***	0.4**	0.6***	0.2	0.4
2	-3.9	-3.6	0.0	-1.2	4.1	0.2	0.2	0.0	-0.1	-0.0
3	0.4	5.7*	7.3**	12.8 ***	21.5***	0.2	-0.4*	-0.1	-0.4	-1.7**
6-month horizon (Diff. from Median)										
Price Tercile	Econ Decline Prob. Spread Quintile					Cumulative Econ growth Spread Quintile				
	1	2	3	4	5	1	2	3	4	5
1	-7.9**	-7.1**	-3.0	2.3	1.7	1.5***	1.1***	0.7**	0.3	0.4
2	-3.0	-2.0	0.0	-0.1	9.3**	0.6	0.1	0.0	-0.1	-0.5
3	1.7	6.5	8.3**	13.5***	24.2**	0.6*	-0.6*	-0.4	-0.6	-2.7*

Table 9: Number of Crashes Preceded by R-zone

Panel A presents the percentage of redzones followed by a market crash within one, three, and six months (PPV), the percentage of market crashes preceded red zones within one, three, and six months (TPR), and the percentage of non-crash month not preceded by an redzone within one, three, and six months (TNR) along with the numbers used for these metrics. Panel B presents the results of an identical analysis with $Y\text{-Zone}_{i,t} \equiv \mathbf{1}\{\Delta_{12}Spread_{i,t} > 60^{th}percentile\} \times \mathbf{1}\{\Delta_{12}log(P)_{i,t} < 33.3^{th}percentile\}$.

Panel A: R-zone			
	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}
#R-zone Events followed by a Crash	165	204	210
#R-zone Events	1,170	1,170	1,170
%R-zone Events followed by a Crash(PPV)	14.1	17.4	18.0
#Crashes Preceded By R-zone	165	223	233
#Crashes	396	411	413
%of Crashes preceded by R-zone(TPR)	41.7	54.3	56.4
#Non-Crashes not Preceded By R-zone	8,198	7,843	7,371
#Non-Crashes	9,193	9,108	9,001
% of Non-Crashes not preceded by R-zone(TNR)	89.2	86.1	81.9
Time to Crash	7.3	7.1	6.7
Panel B: Y-zone			
	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}
#R-zone Events followed by a Crash	188	231	231
#R-zone Events	1,712	1,712	1,712
%R-zone Events followed by a Crash(PPV)	11.0	13.5	13.5
#Crashes Preceded By R-zone	188	254	257
#Crashes	396	411	413
%of Crashes preceded by R-zone(TPR)	47.5	61.8	62.2
#Non-Crashes not Preceded By R-zone	7,713	7,220	6,599
#Non-Crashes	9,193	9,108	9,001
% of Non-Crashes not preceded by R-zone(TNR)	83.9	79.3	73.3
Time to Crash	7.7	7.6	7.1

Table 10: Econ Growth Following True and False Positives

This table presents results of the regression model

$$\Delta \log(IP)_{i, t+1 \text{ to } t+h} = \alpha_i^h + \gamma^{h, tp} \cdot R\text{-Zone}_{i,t} \cdot Crash_{i, t+1 \text{ to } t+6} + \gamma^{h, fp} \cdot R\text{-Zone}_{i,t} \cdot (1 - Crash_{i, t+1 \text{ to } t+6}) + \epsilon_i^h$$

$h = 1, 3, \text{ and } 6$, where $\Delta \log(IP)_{i, t+1 \text{ to } t+h}$ is the log IP growth in country i from month $t + 1$ to month $t + h$. $Crash_{i, t+1 \text{ to } t+h}$ is defined as above. t-statistics are reported in brackets and based on Driscoll and Kraay (1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively, using Kiefer and Vogelsang(2005) corrected p-values. Reported coefficients and R^2 s are in percent.

	1m log econ growth	3m log econ growth	6m log econ growth
True Positives(γ^{tp})	-0.9*** [-2.98]	-3.1*** [-3.48]	-7.9*** [-3.40]
False Positives(γ^{fp})	-0.7*** [-2.79]	-1.6* [-1.97]	-1.9* [-1.91]
$R^2(\text{within})$	0.6	2.5	5.7

Table 11: Portfolio Flows along Market Crashes

This table presents results of the regression model

$$Fund-Flow_{i, t+1 \text{ to } t+h} = \alpha_i^h + \beta^h \cdot Widened-Spread_{i,t} + \delta^h \cdot Weak-Performance_{i,t} + \gamma^h \cdot R-Zone_{i,t} + \epsilon_i^h$$

$$Fund-Flow_{i, t+1 \text{ to } t+h} = \alpha_i^h + \gamma^{h,tp} \cdot R-Zone_{i,t} \cdot Crash_{i, t+1 \text{ to } t+6} + \gamma^{h,fp} \cdot R-Zone_{i,t} \cdot (1 - Crash_{i, t+1 \text{ to } t+6}) + \epsilon_i^h$$

$h = 1, 3$, and 6 , where $Fund-Flow_{i, t+1 \text{ to } t+h}$ is the change of reported portfolio investment assets in country i from month $t + 1$ to month $t + h$. $Crash_{i, t+1 \text{ to } t+h}$ is defined as above. t-statistics are reported in brackets and based on Driscoll and Kraay (1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively, using Kiefer and Vogelsang(2005) corrected p-values. Reported coefficients and R^2 s are in percent.

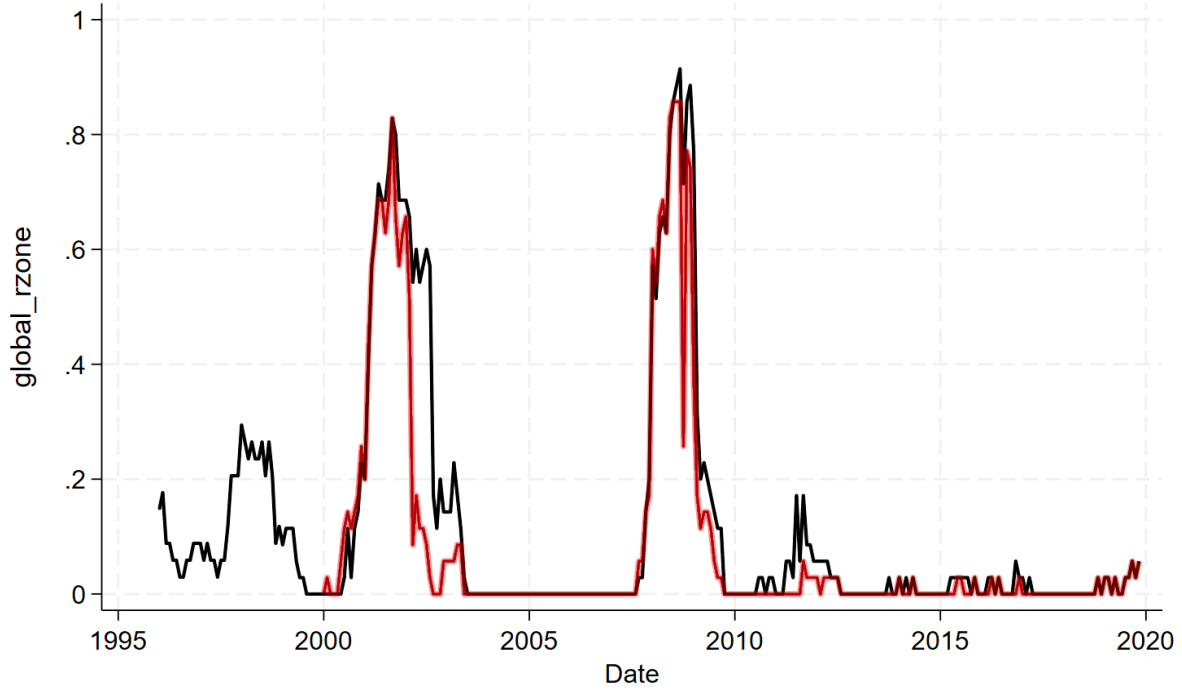
	Fund Flows	Fund Flows	Fund Flows
<i>Widened-Spread</i> (β)	-0.1**	-0.1**	-
	[-2.52]	[-2.25]	-
<i>Weak-Performance</i> (δ)	0.0	0.0	-
	[0.52]	[0.50]	-
<i>R-Zone</i>	-	0.0	-
	-	[0.10]	-
True Positives(γ^{tp})	-	-	-0.3***
	-	-	[-9.73]
False Positives(γ^{fp})	-	-	0.1*
	-	-	[1.89]
$R^2(within)$	0.9	0.9	4.3

Fig. 1. Fraction of Countries in R-zone

The figure depicts the fraction of countries in the red zone at a given time,

$$Global\ R-Zone_t \equiv \frac{1}{N_t} \sum_{i \in S_t} R-Zone_{i,t}$$

where N_t is the number of countries in our sample at time t and S_t is the set of countries in the sample at time t . The dark line represents the Global R-zone using the full sample, while the red line represents the Global R-zone using the expanding sample.



Appendix A. Interest Rate Differentials Decompositions: Local vs US Short-term Interest rate changes

Table A1: Linear Regression - Returns

This table presents results of the regression model:

$$Ret_{i, t+1 \text{ to } t+h} = \alpha_i^h + \beta^h \cdot \Delta_{12}X_{i,t} + \epsilon_{i, t+1 \text{ to } t+h}$$

$h = 1, 3$, and 6 , where α_i^h is a country fixed effect, Δ_{12} is the change in predictor $X_{i,t}$ over the twelve months ending in t , and $Ret_{i, t+1 \text{ to } t+h}$ is stock returns in country i , spanning from month $t + 1$ to month $t + h$. t-statistics are reported in brackets and based on Driscoll and Kraay (1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively, using Kiefer and Vogelsang(2005) corrected p-values. Reported coefficients and R^2 s are in percent.

	Ret _{t+1}	Ret _{t+3}	Ret _{t+6}	Ret _{t+1}	Ret _{t+3}	Ret _{t+6}
$\Delta_{12}st_{Local}$	-0.4*** [-2.81]	-1.4*** [-2.86]	-2.5*** [-2.78]			
$-\Delta_{12}st_{US}$				-0.6* [-1.90]	-1.8 [-1.48]	-3.2 [-1.52]
$R^2(within)$	0.4	1.3	1.9	0.8	2.1	3.3
N	10,009	10,009	10,009	10,009	10,009	10,009

Table A2: Linear Regression - Crashes

This table presents results of the regression model:

$$Crash_{i, t+1 \text{ to } t+h} = \alpha_i^h + \beta^h \cdot \Delta_{12}X_{i,t} + \epsilon_{i, t+1 \text{ to } t+h}$$

$h = 1, 3$, and 6 , where α_i^h is a country fixed effect, Δ_{12} is the change in predictor $X_{i,t}$ over the twelve months ending in t , and $Crash_{i, t+1 \text{ to } t+h}$ is an indicator variable equal to one if a crash begins in country i in month between $t + 1$ and month $t + h$. t-statistics are reported in brackets and based on Driscoll and Kraay (1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively, using Kiefer and Vogelsang(2005) corrected p-values. Reported coefficients and R^2 s are in percent.

	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}
$\Delta_{12}st_{Local}$	0.4 [1.15]	1.2** [2.24]	1.2* [1.67]			
$-\Delta_{12}st_{US}$				2.0** [2.51]	3.1* [1.82]	3.8 [1.50]
$R^2(within)$	0.0	0.4	0.4	1.1	2.5	3.7
N	10,009	10,009	10,009	10,009	10,009	10,009

Table A3: Crashes Prediction with Interest Rate Changes

This table presents results of the regression model

$$Crash_{i, t+1 \text{ to } t+h} = \alpha_i^h + \beta^h \cdot High-\Delta_{12}st_{i,t} + \epsilon_{i, t+1 \text{ to } t+h}$$

$h = 1, 3, \text{ and } 6$. $High-\Delta_{12}st_{i,t} \equiv \mathbf{1}\{\Delta_{12}st_{i,t} > 80^{th} \text{percentile}\}$ is an indicator variable that takes the value of one if twelve-month short-term interest rate changes in the highest quintile. t-statistics are reported in brackets and based on Driscoll and Kraay (1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively, using Kiefer and Vogelsang(2005) corrected p-values. Reported coefficients and R^2 s are in percent.

	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}
$High-\Delta_{12}st_{Local}(\beta^{Local})$	2.6** [2.12]	4.8** [2.44]	6.5* [1.89]			
$High-\Delta_{12}st_{US}(\beta^{US})$				4.8** [2.07]	7.0 [1.54]	9.3 [1.44]
$R^2(\text{within})$	0.3	0.9	1.7	1.0	2.0	3.5
N	10,009	10,009	10,009	10,009	10,009	10,009

Table A4: Crashes Prediction with Global R-zones

This table presents results of the regression model

$$Crash_{i, t+1 \text{ to } t+h} = \alpha_i^h + \gamma^h \cdot Local\ R\text{-}Zone_{i,t} + \xi^h \cdot Global\ R\text{-}Zone_t + \epsilon_{i, t+1 \text{ to } t+h}$$

$h = 1, 3, \text{ and } 6$. $Global\ R\text{-}Zone_t$ measures the fraction of countries in the red zone at a given month. t-statistics are reported in brackets and based on Driscoll and Kraay (1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively, using Kiefer and Vogelsang(2005) corrected p-values. Reported coefficients and R^2 s are in percent.

Panel A						
	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}
Local $R\text{-}Zone$ ($\Delta_{12}st_{Local}$)	3.8** [2.09]	5.1** [2.31]	4.3** [1.97]			
Global $R\text{-}Zone$ ($\Delta_{12}st_{Local}$)	26.3*** [2.72]	39.0** [2.00]	64.3** [2.46]			
Local $R\text{-}Zone$ ($-\Delta_{12}st_{US}$)				3.7* [1.92]	1.73 [0.73]	2.1 [0.59]
Global $R\text{-}Zone$ ($-\Delta_{12}st_{US}$)				5.1 [1.30]	10.4 [1.26]	11.7 [0.99]
$R^2(within)$	3.0	6.3	14.5	2.0	3.4	4.4
N	10,009	10,009	10,009	10,009	10,009	10,009
Panel B						
	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}
Global $R\text{-}Zone$ ($\Delta_{12}st_{Local}$)	30.0*** [3.04]	44.1** [2.26]	68.7*** [2.67]			
Global $R\text{-}Zone$ ($-\Delta_{12}st_{US}$)				8.9** [2.33]	12.1* [1.67]	13.8 [1.49]
$R^2(within)$	2.8	5.9	14.3	1.9	3.4	4.4
N	10,009	10,009	10,009	10,009	10,009	10,009

Appendix B. Alternative Crash Threshold and Horizons

Table A5: Linear Regression - Returns

This table presents results of the regression model:

$$Ret_{i, t+1 \text{ to } t+h} = \alpha_i^h + \beta^h \cdot \Delta_j X_{i,t} + \epsilon_{i, t+1 \text{ to } t+h}$$

$h = 1, 3$, and 6 , where α_i^h is a country fixed effect, Δ_j is the change in predictor $X_{i,t}$ over the six/eighteen months ending in t , and $Ret_{i, t+1 \text{ to } t+h}$ is stock returns in country i , spanning from month $t + 1$ to month $t + h$. t-statistics are reported in brackets and based on Driscoll and Kraay (1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively, using Kiefer and Vogelsang(2005) corrected p-values. Reported coefficients and R^2 s are in percent.

Panel A						
	Ret _{t+1}	Ret _{t+3}	Ret _{t+6}	Ret _{t+1}	Ret _{t+3}	Ret _{t+6}
$\Delta_6 Spread$	-0.3*	-1.7***	-3.6**			
	[-1.88]	[-2.98]	[-2.45]			
$\Delta_6 log(P)$				0.4	0.9	1.4
				[1.29]	[1.05]	[1.00]
$R^2(within)$	0.3	1.9	4.2	0.4	0.5	0.6
N	10,213	10,213	10,213	10,213	10,213	10,213
Panel A						
	Ret _{t+1}	Ret _{t+3}	Ret _{t+6}	Ret _{t+1}	Ret _{t+3}	Ret _{t+6}
$\Delta_{18} Spread$	-0.7***	-2.0**	-3.6**			
	[-2.76]	[-2.17]	[-1.99]			
$\Delta_{18} log(P)$				0.1	0.1	-0.3
				[0.51]	[0.14]	[-0.17]
$R^2(within)$	1.0	2.7	3.9	0.1	0.0	0.0
N	9,805	9,805	9,805	9,805	9,805	9,805

Table A6: Linear Regression - Crashes

This table presents results of the regression model:

$$Crash_{i, t+1 \text{ to } t+h} = \alpha_i^h + \beta^h \cdot \Delta_j X_{i,t} + \epsilon_{i, t+1 \text{ to } t+h}$$

$h = 1, 3$, and 6 , where α_i^h is a country fixed effect, Δ_j is the change in predictor $X_{i,t}$ over the six/eighteen months ending in t , and $Crash_{i, t+1 \text{ to } t+h}$ is an indicator variable equal to one if a crash begins in country i in month between $t + 1$ and month $t + h$. t-statistics are reported in brackets and based on Driscoll and Kraay (1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively, using Kiefer and Vogelsang(2005) corrected p-values. Reported coefficients and R^2 s are in percent.

Panel A						
	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}
$\Delta_6 Spread$	0.8*	1.9***	2.5			
	[1.71]	[2.77]	[1.39]			
$\Delta_6 log(P)$				-2.6***	-2.6***	-2.5**
				[-3.83]	[-2.85]	[-2.08]
$R^2(within)$	0.2	1.0	1.6	1.8	1.8	1.7
N	10,213	10,213	10,213	10,213	10,213	10,213
Panel B						
	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}
$\Delta_{18} Spread$	1.5**	3.2**	3.4			
	[2.06]	[2.25]	[1.60]			
$\Delta_{18} log(P)$				-2.2***	-2.5***	-1.6*
				[-3.37]	[-2.64]	[-1.86]
$R^2(within)$	0.6	2.5	2.8	1.2	1.6	0.6
N	9,805	9,805	9,805	9,805	9,805	9,805

Table A7: Linear Regression - Crashes (Alternative Crash Threshold)

This table presents results of the regression model:

$$Crash_{i, t+1 \text{ to } t+h} = \alpha_i^h + \beta^h \cdot \Delta_{12}X_{i,t} + \epsilon_{i, t+1 \text{ to } t+h}$$

$h = 1, 3$, and 6 , where α_i^h is a country fixed effect, Δ_{12} is the change in predictor $X_{i,t}$ over the twelve months ending in t , and $Crash_{i, t+1 \text{ to } t+h}$ is an indicator variable equal to one if a crash begins in country i in month between $t + 1$ and month $t + h$. t-statistics are reported in brackets and based on Driscoll and Kraay (1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively, using Kiefer and Vogelsang(2005) corrected p-values. Reported coefficients and R^2 s are in percent.

Panel A: Crashes (Ret<5%TH)						
	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}
$\Delta_{12}Spread$	2.1*** [2.80]	3.6** [2.31]	4.23* [1.70]			
$\Delta_{12}log(P)$				-3.1*** [-3.77]	-3.5*** [-2.78]	-3.0** [-2.31]
$R^2(within)$	1.0	2.7	3.7	2.0	2.6	1.9
N	10,009	10,009	10,009	10,009	10,009	10,009
Panel B: Crashes (Ret<3%TH)						
	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}
$\Delta_{12}Spread$	1.5** [2.49]	2.6** [2.03]	3.0 [1.57]			
$\Delta_{12}log(P)$				-2.4*** [-3.63]	-2.6*** [-2.64]	-2.1** [-2.15]
$R^2(within)$	0.7	2.2	3.0	2.0	2.2	1.4
N	10,009	10,009	10,009	10,009	10,009	10,009

Appendix C. Crashes Definition Following Greenwood, Hanson, Shleifer, and Sørensen (2022)

Table A8: Crashes Probabilities by Price and Spread Change Quantiles
Table presents the probability of a crash within one to six months for the intersections of the price terciles and spread change quintiles. It also presents the difference in future crash probability between each group and the median group, which is defined as the intersection of the second price tercile and the third spread change quintile. Both price growth and spread change are measured over twelve months. p-values are based on Driscoll and Kraay(1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively, and corrected according to Kiefer and Vogelsang (2005). *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Crash Probabilities(%) by Spread and Equity Returns										
1-month horizon										
Price Tercile	Crash Frequency Spread Quintile					Diff. from Median Spread Quintile				
	1	2	3	4	5	1	2	3	4	5
1	3.1	1.2	1.8	2.1	4.2	2.5**	0.5	1.1*	1.4	3.5**
2	1.8	0.7	0.7	1.1	5.1	1.1*	0.1	0.0	0.4	4.5**
3	7.8	3.2	3.7	4.2	14.1	7.2***	2.5*	3.0***	3.6***	13.4***
3-month horizon										
Price Tercile	Crash Frequency Spread Quintile					Diff. from Median Spread Quintile				
	1	2	3	4	5	1	2	3	4	5
1	9.4	3.0	4.8	6.6	18.1	7.8***	1.4	3.2***	5.0	16.5***
2	6.3	2.1	1.6	2.8	13.8	4.7**	0.5	0.0	1.2	12.2***
3	18.0	7.1	11.8	11.9	32.0	16.4***	5.6***	10.2***	10.3***	30.4***
6-month horizon										
Price Tercile	Crash Frequency Spread Quintile					Diff. from Median Spread Quintile				
	1	2	3	4	5	1	2	3	4	5
1	17.4	4.5	9.9	14.4	33.4	14.5***	1.6	6.9***	11.4**	30.5***
2	10.3	4.6	3.0	6.4	24.5	7.3*	1.7	0.0	3.4*	21.5***
3	31.0	11.5	15.6	21.8	48.3	28.0***	8.5***	12.7***	18.8***	45.4***

Table A9: Crashes Prediction with Spread Change and Stock Performance
This table presents results of the regression model

$$Crash_{i, t+1 \text{ to } t+h} = \alpha_i^h + \beta^h \cdot Widened-Spread_{i,t} + \delta^h \cdot Weak-Performance_{i,t} + \gamma^h \cdot R-Zone_{i,t} + \epsilon_i^h$$

$h = 1, 3, \text{ and } 6$. $Widened-Spread_{i,t} \equiv \mathbf{1}\{\Delta_{12}Spread_{i,t} > 80^{th}percentile\}$ is an indicator variable that takes the value of one if twelve-month spread change is in the highest quintile, and $Weak-Performance_{i,t} \equiv \mathbf{1}\{\Delta_{12}\log(P)_{i,t} < 33.3^{th}percentile\}$ is an indicator variable that takes the value of one if equity performance is in its lowest tercile, an R-zone is the intersection of high spread change and low stock performance $R-Zone_{i,t} \equiv Widened-Spread_{i,t} \times Weak-Performance_{i,t}$. The sum of the coefficients $\beta^h + \delta^h + \gamma^h$ gives the increase in the probability that a crisis begins within h months when widened spread and weak stock performance are jointly elevated. t-statistics are reported in brackets and based on Driscoll and Kraay (1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively, using Kiefer and Vogelsang(2005) corrected p-values. Reported coefficients and R^2 s are in percent.

	Crash _{t+1}				Crash _{t+3}				Crash _{t+6}			
<i>Widened-Spread</i> (β)	7.8*** [3.72]		3.1* [1.92]		18.6*** [3.72]		11.0*** [2.71]		28.0*** [4.18]		19.2*** [2.99]	
<i>Weak-Performance</i> (δ)		5.6*** [4.12]	2.6*** [3.92]			11.9*** [3.37]	6.0*** [3.40]			16.5*** [3.49]	8.4*** [2.86]	
<i>R-Zone</i> (γ)			6.7** [2.21]	11.3*** [3.73]			9.7 [1.51]	24.1*** [3.74]			10.4* [1.77]	34.1*** [4.90]
Sum of coefficients($\beta + \delta + \gamma$)			12.3***				26.7***				38.1***	
t-statistic($\beta + \delta + \gamma$)			[3.94]				[4.03]				[5.13]	
R^2 (within)	2.5	1.8	3.9	6.1	3.5	8.0	6.7	6.0	9.2	4.4	11.3	8.9

Table A10: Crashes Prediction with R-zones: Robust

This table presents results of the regression model

$$Crash_{i, t+1 \text{ to } t+h} = \alpha_i^h + \gamma^h \cdot R\text{-Zone}_{i,t} + \epsilon_{i, t+1 \text{ to } t+h}$$

$h = 1, 3$, and 6 . An R-zone is the intersection of high spread change and low stock performance $R\text{-Zone}_{i,t} \equiv \mathbf{1}\{\Delta_{12}Spread_{i,t} > 80^{th}percentile\} \times \mathbf{1}\{\Delta_{12}log(P)_{i,t} < 33.3^{th}percentile\}$. Besides examining advanced market and emerging market, we also investigate the expanding sample evidence. In expanding sample specification, R-zone indicators in each month t are based on a expanding sample using past data, that is, the R-zone indicator in 2006m12 is based on data from 1995m1 to 2006m12, and we require at least 60 months of data. t-statistics are reported in brackets and based on Driscoll and Kraay (1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively, using Kiefer and Vogelsang(2005) corrected p-values. Reported coefficients and R^2 s are in percent.

<i>R-Zone</i>	Crash _{$t+1$}			Crash _{$t+3$}			Crash _{$t+6$}		
	ADV	EM	Expanding	ADV	EM	Expanding	ADV	EM	Expanding
	12.0*** [3.44]	10.3*** [3.44]	10.2** [2.43]	28.4*** [3.80]	17.1*** [2.87]	24.3** [2.53]	41.9*** [5.45]	21.4*** [2.93]	38.6*** [3.77]
$R^2(\text{within})$	4.6	2.3	2.5	10.6	2.8	6.0	10.5	3.1	10.1
N	6,565	3,444	8,365	6,519	3,420	8,295	6,450	3,384	8,190

Table A11: Crashes Prediction with Global R-zones

This table presents results of the regression model

$$Crash_{i, t+1 \text{ to } t+h} = \alpha_i^h + \gamma^h \cdot Local\ R\text{-}Zone_{i,t} + \xi^h \cdot Global\ R\text{-}Zone_t + \epsilon_{i, t+1 \text{ to } t+h}$$

$h = 1, 3, \text{ and } 6$. *Global R-Zone_t* measures the fraction of countries in the red zone at a given month. t-statistics are reported in brackets and based on Driscoll and Kraay (1998) standard errors with lags of zero, five, and nine months for prediction horizons one, three, and six months, respectively. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively, using Kiefer and Vogelsang(2005) corrected p-values. Reported coefficients and R^2 s are in percent.

	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}	Crash _{t+1}	Crash _{t+3}	Crash _{t+6}
Local <i>R-Zone</i>	4.3*** [2.80]	6.1** [2.01]	8.0** [2.00]			
Global <i>R-Zone</i>	15.3** [2.55]	39.1*** [3.63]	56.7*** [4.07]	19.7*** [3.28]	45.3*** [4.06]	64.7*** [4.97]
$R^2(\text{within})$	5.1	11.1	15.0	4.8	10.8	14.7
N	10,009	9,939	9,834	10,009	9,939	9,834